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Study vehicle routing problem using Nearest Neighbor Algorithm

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Study vehicle routing problem using Nearest Neighbor Algorithm

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Abstract. Vehicle routing problem (VRP) has a key role in logistics management. VRP plays a role in designing the optimal route used by a number of vehicles placed at the depot to serve a number of customers with known requests. To solve the VRP, the nearest neighbor algorithm used to get the most optimal results. The Nearest Neighbor algorithm is a heuristic method which is done by starting the starting point than looking for the nearest point. In this paper, algorithm nearest neighbour can save the distance of 13,14% and cost 13,17%.

Keywords: optimization, algorithm, distance.

1. Introduction

At present the problem of decision making has a great influence on distribution actors to minimize mileage, time and cost. For example, schools that use school buses to pick up and deliver their students, pizza couriers deliver their products to customers, and container trucks deliver their product from depot to several retail stores, and than distribute goods from warehouses to many retail stores requires several fleets for delivery. It takes the right decision to choose the store that will be visited first so that mileage, time and cost are the minimum.

Vehicle Routing Problem (VRP) is a matter of distributing goods from the company's depot (warehouse) to a number of customers according to their respective requests with the aim of minimizing the total vehicle mileage, vehicle travel time and transportation costs under conditions:

- Each city can only be visited by one vehicle in one route
- The capacity of each vehicle is enough for all vehicles in one route
- Routes start and end in the same city

The types of VRP are VRP Time Windows (VRPTW), VRP Pick-Up and Delivery (VRPPD) and Capacitated VRP (CVRP).



In this paper, the authors used the Nearest Neighbor algorithm because this method is the simplest and has the characteristics of forming distribution routes according to the real conditions in the field. Nearest Neighbor is a heuristic method used to solve VRP. The settlement is done by starting the first point then looking for the closest point. After all locations have passed, the fleet must return to the starting point.

2. Mathematical Formulation the VRP

Let x_{ij} be an integer variable which may take value $\{0, 1\}$, $\forall \{i, j\} \in E$ $\{\{0, j\} : j \in V\}$ and value $\{0, 1, 2\}$ $\forall \{0, j\} \in E, j \in V$. Note that $x_{0j} = 2$ when a route including the single customer j is selected in solution

VRP can be formulated as follows :

$$\min \sum_{i \neq j} d_{ij} x_{ij}$$

Subject to:

$$\sum_j x_{ij} = 1, \forall i \in V,$$

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$$\sum_i x_{ij} |S| - c(S), \{S : S \subseteq V \setminus \{1\}, |S| \geq 2\},$$

$$x_{ij} \in \{0, 1\}, \forall \{i, j\} \in E; i \neq j$$

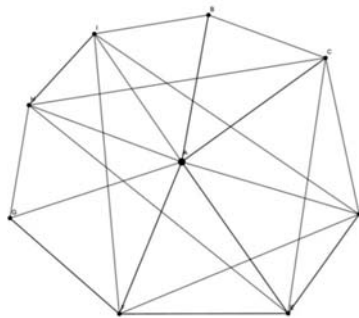


Figure 1. VRP case

Figure 1 shows an example of a VRP case. There are 8 points that will be visited by 2 vehicles to deliver goods with a certain distance between points. Point A as a depot, and then point B,C,D,E,F,G,H,I as the location to be visited

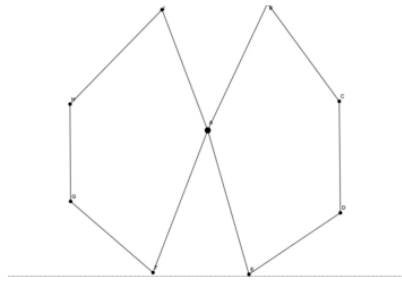


Figure 2. Solving VRP problem

Figure 2 is the solving of VRP where the locations B, C, D, E visited by vehicle 1, and then locations F, G, H, I visited by vehicle 2.

3. Nearest Neighbor Algorithm

The nearest neighbor effective algorithm solves optimization problems as in VRP. The steps are follows:

- Select a point that represents the city or starting location.
- Then select the destination point or city to be visited next, with consideration only choosing the city that has the closest distance to the city that was previously visited.
- If all cities have been visited or all points have been connected, then close the trip route by returning to the city of origin (starting point).

4. Discussion and Result

The data in this paper form of literature study obtained from the essay with the title Ants Algorithm Implementation for Optimization of the Shortest Route written by Ernawati in 2017. The data will be used for this study using the nearest neighbor algorithm as follows:

JNE office will deliver goods to 6 location points with 2 vehicles.

Table 1. Costemers data will be served

No.	Delivery Locations	Code
1	The JNE Office	X
2	Instalasi Kusta dan Infeksi Endemik	A
3	RSIA Siti Khadijah III	B
4	PT. Indolok Bakti	C
5	Dipa Jaya Sejahtera	D
6	Mal GTC Tanjung Bunga	E
7	Dinas Kelautan dan Perikanan	F

The distances between locations are listed in the table below:

Table 2. Distance and time of JNE delivery locations

No.	Location	Distance in Km	Period in minutes
1	X-B	2.5 km	9 m
2	X-B	2.9 km	8 m
3	X-C	2.9 km	8 m
4	X-D	3.3 km	9 m
5	X-E	6.2 km	16 m
6	X-F	2.3 km	8 m
7	A-B	1.1 km	4 m
8	A-C	1.3 km	4 m
9	A-D	1.9 km	6 m
10	A-E	6.7 km	16 m
11	A-F	2.9 km	9 m
12	B-C	0.8 km	2 m
13	B-D	1.7 km	6 m
14	B-E	8.1 km	19 m
15	B-F	3 km	8 m
16	C-D	1.2 km	4 m
17	C-E	5.4 km	13 m
18	C-F	2.4 km	7 m
19	D-E	5.6 km	11 m
20	D-F	2.4 km	7 m
21	E-F	3.4 km	8 m

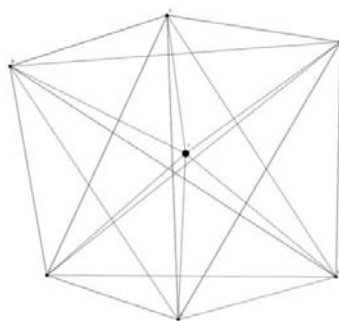


Figure 3. VRP case

Figure 3 shows X as a JNE office, then A, B, C, D E, F as the locations will be visited by two couriers.

Table 3. The parameters of vehicles

No.	Fleets	Merk brand	Usage of pertamax per liter	Price per liter	Price per km
1	Courier 1	Vario 150 cc	52.9 km/liter	Rp 9.850,-/liter	Rp 186.2,-/km
2	Courier 2	Vario 150 cc	52.9 km/liter	Rp 9.850,-/liter	Rp 186.2,-/km

Fuel usage and prices data is taken from <https://www.hondacengkareng.com/daftarkonsumsi-bbm-sepeda-motor-honda/> and <https://www.pertamina.com/id/newsroom/announcement/daftar-harga-bbk-tmt-10-februari-2019>

4.1. VRP processing before using nearest neighbor algorithm

Without nearest neighbor algorithm, author solves the VRP problem.

The location visited Courier 1 = $X \rightarrow E \rightarrow D \rightarrow A \rightarrow X$

The distance traveled is 6,2 km + 5,6 km + 1,9 km + 2,5 km. The totals distance is 16,2 km. Courier 1 travel time is 16 minute + 11 minute + 6 minute + 9 minute. The totals time taken by courier 1 from point X to back point X is 42 minutes. The cost for fuel used courier 1 is 16,2 km $\times$ Rp 186,2. Totals cost is Rp 3.016,44.

The location visited Courier 2 = $X \rightarrow C \rightarrow F \rightarrow B \rightarrow X$

The distance traveled is 2,9 km + 2,4 km + 3 km + 2,9 km. The totals distance is 11,2 km. Courier 2 travel time is 2 adalah 8 minute + 7 minute + 8 minute + 8 minute. The totals time taken by courier 1 from point X to back point X is 31 minutes. The cost for fuel used courier 1 is 11,2 km $\times$ Rp 186,2. Totals cost is Rp 2.085,44.

The total distance taken by the two couriers to deliver goods from the JNE office and back to the JNE office is 16.2 km + 11.2 km = 27.4 km. The longest total travel time is 42 minutes. Then the costs incurred by the JNE Office for the fuel of the two couriers are Rp. 3,016.44 + Rp. 2,085.44 = Rp. 5,101.88.

4.2. VRP using nearest neighbor algorithm

VRP settlement using nearest neighbor algorithm.

The location visited courier 1 from X location is: first look at the nearest location from point X. In table 1, nearest location from point X is A. From point A, the courier goes to point B. From point B, the courier goes to point C and go back to point X. The distance traveled is 2,5 km + 1,1 km + 0,8 km + 2,9 km. The totals distance is 7,3 km. The totals time taken by courier 1 from point X to back point X is 23 minutes. The cost for fuel used courier 1 is 7,3 km $\times$ Rp 186,2. Totals cost is Rp 1.357,8.

The total distance taken by the two couriers to deliver goods from the JNE office and back to the JNE office is 7,3 km + 16,5 km = 23,8 km. The longest total travel time is 42 minutes. Then the costs incurred by the JNE Office for the fuel of the two couriers are Rp. 1.357,80 + Rp. 3.072,30 = Rp. 4.430,10.

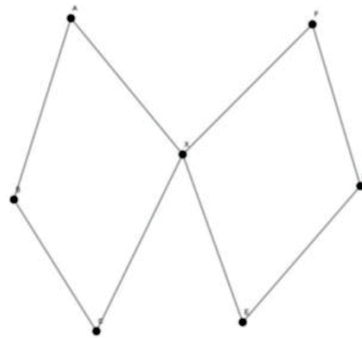


Figure 4. solve using nearest neighbor

5. Conclusion

Based on data analysis before and after using the nearest neighbor algorithm, it can be concluded:

- The nearest neighbor algorithm can be solve vrp problems and the results obtained are more optimal.
- The total distance traveled by the courier obtained a distance saving of 3.6 km or 13,14%
- The total cost by the courier obtained a cost saving Rp 671,78 or 13,17 %.
- Nearest neighbor algorithm can be used as a reference for the JNE Office to minimize distance and cost.

6. References

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